

Spaces with Polynomial Cohomology

We saw in Theorem 3.12 that $\mathbb{R}P^\infty$, $\mathbb{C}P^\infty$, and $\mathbb{H}P^\infty$ have cohomology rings that are polynomial algebras. We will describe now a construction for enlarging S^{2n} to a space $J(S^{2n})$ whose cohomology ring $H^*(J(S^{2n}); \mathbb{Z})$ is almost the polynomial ring $\mathbb{Z}[x]$ on a generator x of dimension $2n$. And if we change from $\mathbb{Z}$ to $\mathbb{Q}$ coefficients, then $H^*(J(S^{2n}); \mathbb{Q})$ is exactly the polynomial ring $\mathbb{Q}[x]$. This construction, known as the **James reduced product**, is also of interest because of its connections with loopspaces described in §4.J.

For a space X , let X^k be the product of k copies of X . From the disjoint union $\coprod_{k \geq 1} X^k$, let us form a quotient space $J(X)$ by identifying $(x_1, \dots, x_i, \dots, x_k)$ with $(x_1, \dots, \hat{x}_i, \dots, x_k)$ if $x_i = e$, a chosen basepoint of X . Points of $J(X)$ can thus be thought of as k -tuples $(x_1, \dots, x_k)$, $k \geq 0$, with no $x_i = e$. Inside $J(X)$ is the subspace $J_m(X)$ consisting of the points $(x_1, \dots, x_k)$ with $k \leq m$. This can be viewed as a quotient space of X^m under the identifications $(x_1, \dots, x_i, e, \dots, x_m) \sim (x_1, \dots, e, x_i, \dots, x_m)$. For example, $J_1(X) = X$ and $J_2(X) = X \times X / (x, e) \sim (e, x)$. If X is a CW complex with e a 0-cell, the quotient map $X^m \rightarrow J_m(X)$ glues together the m subcomplexes of the product complex X^m where one coordinate is e . These glueings are by homeomorphisms taking cells onto cells, so $J_m(X)$ inherits a CW structure from X^m . There are natural inclusions $J_m(X) \subset J_{m+1}(X)$ as subcomplexes, and $J(X)$ is the union of these subcomplexes, hence is also a CW complex.

Proposition 3.22. *For $n > 0$, $H^*(J(S^n); \mathbb{Z})$ consists of a $\mathbb{Z}$ in each dimension a multiple of n . If n is even, the i^{th} power of a generator of $H^n(J(S^n); \mathbb{Z})$ is $i!$ times a generator of $H^{in}(J(S^n); \mathbb{Z})$, for each $i \geq 1$.*

Thus for n even, $H^*(J(S^n); \mathbb{Z})$ can be identified with the subring of the polynomial ring $\mathbb{Q}[x]$ additively generated by the monomials $x^i/i!$. This subring is called a **divided polynomial algebra** and is denoted $\Gamma_{\mathbb{Z}}[x]$. An exercise at the end of the section is to show that when n is odd, $H^*(J(S^n); \mathbb{Z})$ is isomorphic as a graded ring to $H^*(S^n; \mathbb{Z}) \otimes H^*(J(S^{2n}); \mathbb{Z})$, the tensor product of an exterior algebra and a divided polynomial algebra.

Proof: Giving S^n its usual CW structure, the resulting CW structure on $J(S^n)$ consists of exactly one cell in each dimension a multiple of n . Thus if $n > 1$ we deduce immediately from cellular cohomology that $H^*(J(S^n); \mathbb{Z})$ consists exactly of $\mathbb{Z}$'s in dimensions a multiple of n . An alternative argument that works also when $n = 1$ is the following. Consider the quotient map $q: (S^n)^m \rightarrow J_m(S^n)$. This carries each cell of $(S^n)^m$ homeomorphically onto a cell of $J_m(S^n)$. In particular q is a cellular map, taking k -skeleton to k -skeleton for each k , so q induces a chain map of cellular chain complexes. This chain map is surjective since each cell of $J_m(S^n)$ is the homeomorphic image of a cell of $(S^n)^m$. Hence all the cellular boundary maps for $J_m(S^n)$

will be trivial if the same is true for $(S^n)^m$, which it must be in order for $H^*((S^n)^m; \mathbb{Z})$ to have the structure given by Theorem 3.16.

Since q maps each of the m n -cells of $(S^n)^m$ homeomorphically onto the n -cell of $J_m(S^n)$, we see from cellular cohomology that a generator $\alpha \in H^n(J_m(S^n); \mathbb{Z})$ pulls back by q^* to the sum $\alpha_1 + \cdots + \alpha_m$ of the generators of $H^n((S^n)^m; \mathbb{Z})$ corresponding to the n -cells of $(S^n)^m$. If n is even, the cup product structure in $H^*(J(S^n); \mathbb{Z})$ is strictly commutative and $H^*((S^n)^m; \mathbb{Z}) \approx \mathbb{Z}[\alpha_1, \dots, \alpha_m]/(\alpha_1^2, \dots, \alpha_m^2)$. The power α^m then pulls back to $(\alpha_1 + \cdots + \alpha_m)^m = m! \alpha_1 \cdots \alpha_m$, where the product $\alpha_1 \cdots \alpha_m$ generates $H^{mn}((S^n)^m; \mathbb{Z}) \approx \mathbb{Z}$. The map q is a homeomorphism from the mn -cell of $(S^n)^m$ to the mn -cell of $J_m(S^n)$, so q^* is an isomorphism on H^{mn} . This implies that α^m is $m!$ times a generator of $H^{mn}(J_m(S^n); \mathbb{Z})$. Since the cells of $J(S^n) - J_m(S^n)$ have dimension at least $(m+1)n$, the inclusion $J_m(S^n) \subset J(S^n)$ induces isomorphisms on H^i for $i \leq mn$. Thus if we let x_i denote a generator of $H^{in}(J(S^n); \mathbb{Z})$, we have $x_1^m = \pm m! x_m$ for all m . The sign can be made $+$ by rechoosing x_m if need be. $\square$

In $\Gamma_{\mathbb{Z}}[x] \subset \mathbb{Q}[x]$, if we let $x_i = x^i/i!$ then the multiplicative structure is given by $x_i x_j = \binom{i+j}{i} x^{i+j}$. More generally, for a commutative ring R we could define $\Gamma_R[x]$ to be the free R -module with basis $x_0 = 1, x_1, x_2, \dots$ and multiplication defined by $x_i x_j = \binom{i+j}{i} x^{i+j}$. The preceding proposition implies that $H^*(J(S^{2n}); R) \approx \Gamma_R[x]$. When $R = \mathbb{Q}$ it is clear that $\Gamma_{\mathbb{Q}}[x]$ is just $\mathbb{Q}[x]$. However, for $R = \mathbb{Z}_p$ with p prime something quite different happens: There is an isomorphism

$$\Gamma_{\mathbb{Z}_p}[x] \approx \mathbb{Z}_p[x_1, x_p, x_{p^2}, \dots]/(x_1^p, x_p^p, x_{p^2}^p, \dots) = \bigotimes_{i \geq 0} \mathbb{Z}_p[x_{p^i}]/(x_{p^i}^p)$$

as we show in §3.C, where we will also see that divided polynomial algebras are in a certain sense dual to polynomial algebras.

The examples of projective spaces lead naturally to the following question: Given a coefficient ring R and an integer $d > 0$, is there a space X having $H^*(X; R) \approx R[\alpha]$ with $|\alpha| = d$? Historically, it took major advances in the theory to answer this simple-

looking question. Here is a table giving all the possible values of d for some of the most obvious and important choices of R , namely $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{Z}_2$, and $\mathbb{Z}_p$ with p an odd prime. As we have seen, projective

R	d
$\mathbb{Z}$	2, 4
$\mathbb{Q}$	any even number
$\mathbb{Z}_2$	1, 2, 4
$\mathbb{Z}_p$	any even divisor of $2(p-1)$

spaces give the examples for $\mathbb{Z}$ and $\mathbb{Z}_2$. Examples for $\mathbb{Q}$ are the spaces $J(S^d)$, and examples for $\mathbb{Z}_p$ are constructed in §3.G. Showing that no other d 's are possible takes considerably more work. The fact that d must be even when $R \neq \mathbb{Z}_2$ is a consequence of the commutativity property of cup product. In Theorem 4L.9 and Corollary 4L.10 we will settle the case $R = \mathbb{Z}$ and show that d must be a power of 2 for $R = \mathbb{Z}_2$ and a power of p times an even divisor of $2(p-1)$ for $R = \mathbb{Z}_p$, p odd. Ruling out the remaining cases is best done using K-theory, as in [VBKT] or the classical reference

[Adams & Atiyah 1966]. However there is one slightly anomalous case, $R = \mathbb{Z}_2$, $d = 8$, which must be treated by special arguments; see [Toda 1963].

It is an interesting fact that for each even d there exists a CW complex X_d which is simultaneously an example for all the admissible choices of coefficients R in the table. Moreover, X_d can be chosen to have the simplest CW structure consistent with its cohomology, namely a single cell in each dimension a multiple of d . For example, we may take $X_2 = \mathbb{CP}^\infty$ and $X_4 = \mathbb{HP}^\infty$. The next space X_6 would have $H^*(X_6; \mathbb{Z}_p) \approx \mathbb{Z}_p[\alpha]$ for $p = 7, 13, 19, 31, \dots$, primes of the form $3s + 1$, the condition $6 \mid 2(p - 1)$ being equivalent to $p = 3s + 1$. (By a famous theorem of Dirichlet there are infinitely many primes in any such arithmetic progression.) Note that, in terms of $\mathbb{Z}$ coefficients, X_d must have the property that for a generator α of $H^d(X_d; \mathbb{Z})$, each power α^i is an integer a_i times a generator of $H^{di}(X_d; \mathbb{Z})$, with $a_i \neq 0$ if $H^*(X_d; \mathbb{Q}) \approx \mathbb{Q}[\alpha]$ and a_i relatively prime to p if $H^*(X_d; \mathbb{Z}_p) \approx \mathbb{Z}_p[\alpha]$. A construction of X_d is given in [SSAT], or in the original source [Hoffman & Porter 1973].

One might also ask about realizing the truncated polynomial ring $R[\alpha]/(\alpha^{n+1})$, in view of the examples provided by $\mathbb{RP}^n$, $\mathbb{CP}^n$, and $\mathbb{HP}^n$, leaving aside the trivial case $n = 1$ where spheres provide examples. The analysis for polynomial rings also settles which truncated polynomial rings are realizable; there are just a few more than for the full polynomial rings.

There is also the question of realizing polynomial rings $R[\alpha_1, \dots, \alpha_n]$ with generators α_i in specified dimensions d_i . Since $R[\alpha_1, \dots, \alpha_m] \otimes_R R[\beta_1, \dots, \beta_n]$ is equal to $R[\alpha_1, \dots, \alpha_m, \beta_1, \dots, \beta_n]$, the product of two spaces with polynomial cohomology is again a space with polynomial cohomology, assuming the number of polynomial generators is finite in each dimension. For example, the n -fold product $(\mathbb{CP}^\infty)^n$ has $H^*((\mathbb{CP}^\infty)^n; \mathbb{Z}) \approx \mathbb{Z}[\alpha_1, \dots, \alpha_n]$ with each α_i 2-dimensional. Similarly, products of the spaces $J(S^{d_i})$ realize all choices of even d_i 's with $\mathbb{Q}$ coefficients.

However, with $\mathbb{Z}$ and $\mathbb{Z}_p$ coefficients, products of one-variable examples do not exhaust all the possibilities. As we show in §4.D, there are three other basic examples with $\mathbb{Z}$ coefficients:

1. Generalizing the space $\mathbb{CP}^\infty$ of complex lines through the origin in $\mathbb{C}^\infty$, there is the **Grassmann manifold** $G_n(\mathbb{C}^\infty)$ of n -dimensional vector subspaces of $\mathbb{C}^\infty$, and this has $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z}) \approx \mathbb{Z}[\alpha_1, \dots, \alpha_n]$ with $|\alpha_i| = 2i$. This space is also known as $BU(n)$, the 'classifying space' of the unitary group $U(n)$. It is central to the study of vector bundles and K-theory.
2. Replacing $\mathbb{C}$ by $\mathbb{H}$, there is the quaternionic Grassmann manifold $G_n(\mathbb{H}^\infty)$, also known as $BSp(n)$, the classifying space for the symplectic group $Sp(n)$, with $H^*(G_n(\mathbb{H}^\infty); \mathbb{Z}) \approx \mathbb{Z}[\alpha_1, \dots, \alpha_n]$ with $|\alpha_i| = 4i$.
3. There is a classifying space $BSU(n)$ for the special unitary group $SU(n)$, whose cohomology is the same as for $BU(n)$ but with the first generator α_1 omitted, so $H^*(BSU(n); \mathbb{Z}) \approx \mathbb{Z}[\alpha_2, \dots, \alpha_n]$ with $|\alpha_i| = 2i$.

These examples and their products account for all the realizable polynomial cup product rings with $\mathbb{Z}$ coefficients, according to a theorem in [Adams & Wilkerson 1980]. The situation for $\mathbb{Z}_p$ coefficients is more complicated and will be discussed in §3.G.

Here is the evident general question along these lines:

The Realization Problem. Which graded commutative R -algebras occur as cup product algebras $H^*(X; R)$ of spaces X ?

This is a difficult problem, with the degree of difficulty depending strongly on the coefficient ring R . The most accessible case is $R = \mathbb{Q}$, where essentially every graded commutative $\mathbb{Q}$ -algebra is realizable, as shown in [Quillen 1969]. Next in order of difficulty is $R = \mathbb{Z}_p$ with p prime. This is much harder than the case of $\mathbb{Q}$, and only partial results, obtained with much labor, are known, mainly about realizing polynomial rings. Finally there is $R = \mathbb{Z}$, about which very little is known beyond what is implied by the $\mathbb{Z}_p$ cases.

Polynomial algebras are examples of free graded commutative algebras, where ‘free’ means loosely ‘having no unnecessary relations.’ In general, a free graded commutative algebra is a tensor product of single-generator free graded commutative algebras. The latter are either polynomial algebras $R[\alpha]$ on even-dimension generators α or quotients $R[\alpha]/(2\alpha^2)$ with α odd-dimensional. Note that if R is a field then $R[\alpha]/(2\alpha^2)$ is either the exterior algebra $\Lambda_R[\alpha]$ if the characteristic of R is not 2, or the polynomial algebra $R[\alpha]$ otherwise. Every graded commutative algebra is a quotient of a free one, clearly.

Example 3.23: Subcomplexes of the n -Torus. To give just a small hint of the endless variety of nonfree cup product algebras that can be realized, consider subcomplexes of the n -torus T^n , the product of n copies of S^1 . Here we give S^1 its standard minimal cell structure and T^n the resulting product cell structure. We know that $H^*(T^n; \mathbb{Z})$ is the exterior algebra $\Lambda_{\mathbb{Z}}[\alpha_1, \dots, \alpha_n]$, with the monomial $\alpha_{i_1} \cdots \alpha_{i_k}$ corresponding via cellular cohomology to the k -cell $e_{i_1}^1 \times \cdots \times e_{i_k}^1$. So if we pass to a subcomplex $X \subset T^n$ by omitting certain cells, then $H^*(X; \mathbb{Z})$ is the quotient of $\Lambda_{\mathbb{Z}}[\alpha_1, \dots, \alpha_n]$ obtained by setting the monomials corresponding to the omitted cells equal to zero. Since we are dealing with rings, we are factoring out by an ideal in $\Lambda_{\mathbb{Z}}[\alpha_1, \dots, \alpha_n]$, the ideal generated by the monomials corresponding to the ‘minimal’ omitted cells, those whose boundary is entirely contained in X . For example, if we take X to be the subcomplex of T^3 obtained by deleting the cells $e_1^1 \times e_2^1 \times e_3^1$ and $e_2^1 \times e_3^1$, then $H^*(X; \mathbb{Z}) \approx \Lambda_{\mathbb{Z}}[\alpha_1, \alpha_2, \alpha_3]/(\alpha_2\alpha_3)$.

How many different subcomplexes of T^n are there? To each subcomplex $X \subset T^n$ we can associate a finite simplicial complex C_X by the following procedure. View T^n as the quotient of the n -cube $I^n = [0, 1]^n \subset \mathbb{R}^n$ obtained by identifying opposite faces. If we intersect I^n with the hyperplane $x_1 + \cdots + x_n = \varepsilon$ for small $\varepsilon > 0$, we get a simplex Δ^{n-1} . Then for $q: I^n \rightarrow T^n$ the quotient map, we take C_X to be

$\Delta^{n-1} \cap q^{-1}(X)$. This is a subcomplex of Δ^{n-1} whose k -simplices correspond exactly to the $(k+1)$ -cells of X . Clearly X is uniquely determined by C_X , and it is easy to see that every subcomplex of Δ^{n-1} occurs as C_X for some subcomplex X of T^n . Since every simplicial complex with n vertices is a subcomplex of Δ^{n-1} , we see that T^n has quite a large number of subcomplexes, if n is not too small. Of course, it may be that some of the resulting cohomology rings $H^*(X; \mathbb{Z})$ are isomorphic for different subcomplexes $X \subset T^n$. For example, one could just permute the factors of T^n to change X without affecting its cohomology ring. Whether there are less trivial examples is a harder algebraic problem.

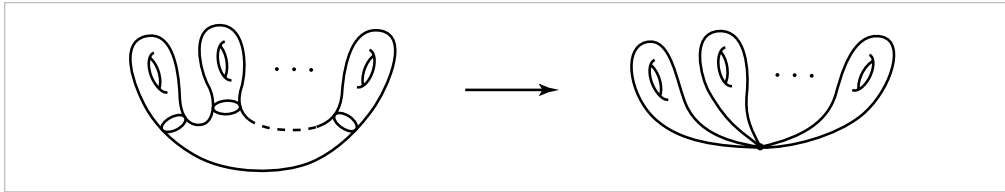
Somewhat more elaborate examples could be produced by looking at subcomplexes of the product of n copies of $\mathbb{CP}^\infty$. In this case the cohomology rings are isomorphic to polynomial rings modulo ideals generated by monomials. One could also take subcomplexes of a product of S^1 's and $\mathbb{CP}^\infty$'s. However, this is still a whole lot less complicated than the general case, where one takes free algebras modulo ideals generated by arbitrary polynomials having all their terms of the same dimension.

Let us conclude this section with an example of a cohomology ring that is not too far removed from a polynomial ring.

Example 3.24: Cohen-Macaulay Rings. Let X be the quotient space $\mathbb{CP}^\infty / \mathbb{CP}^{n-1}$. The quotient map $\mathbb{CP}^\infty \rightarrow X$ induces an injection $H^*(X; \mathbb{Z}) \rightarrow H^*(\mathbb{CP}^\infty; \mathbb{Z})$ embedding $H^*(X; \mathbb{Z})$ in $\mathbb{Z}[\alpha]$ as the subring generated by $1, \alpha^n, \alpha^{n+1}, \dots$. If we view this subring as a module over $\mathbb{Z}[\alpha^n]$, it is free with basis $\{1, \alpha^{n+1}, \alpha^{n+2}, \dots, \alpha^{2n-1}\}$. Thus $H^*(X; \mathbb{Z})$ is an example of a *Cohen-Macaulay ring*: a ring containing a polynomial subring over which it is a finitely generated free module. While polynomial cup product rings are rather rare, Cohen-Macaulay cup product rings occur much more frequently.

Exercises

1. Assuming as known the cup product structure on the torus $S^1 \times S^1$, compute the cup product structure in $H^*(M_g)$ for M_g the closed orientable surface of genus g by using the quotient map from M_g to a wedge sum of g tori, shown below.



2. Using the cup product $H^k(X, A; R) \times H^\ell(X, B; R) \rightarrow H^{k+\ell}(X, A \cup B; R)$, show that if X is the union of contractible open subsets A and B , then all cup products of positive-dimensional classes in $H^*(X; R)$ are zero. This applies in particular if X is a suspension. Generalize to the situation that X is the union of n contractible open subsets, to show that all n -fold cup products of positive-dimensional classes are zero.